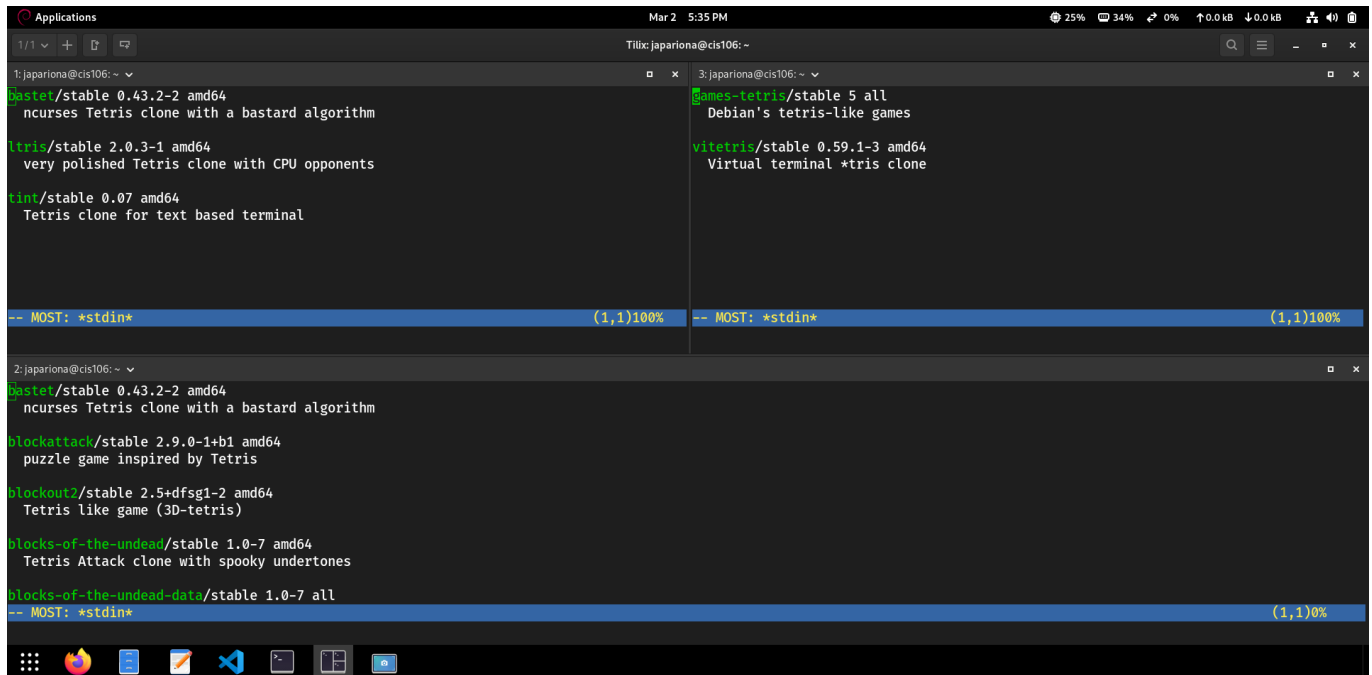


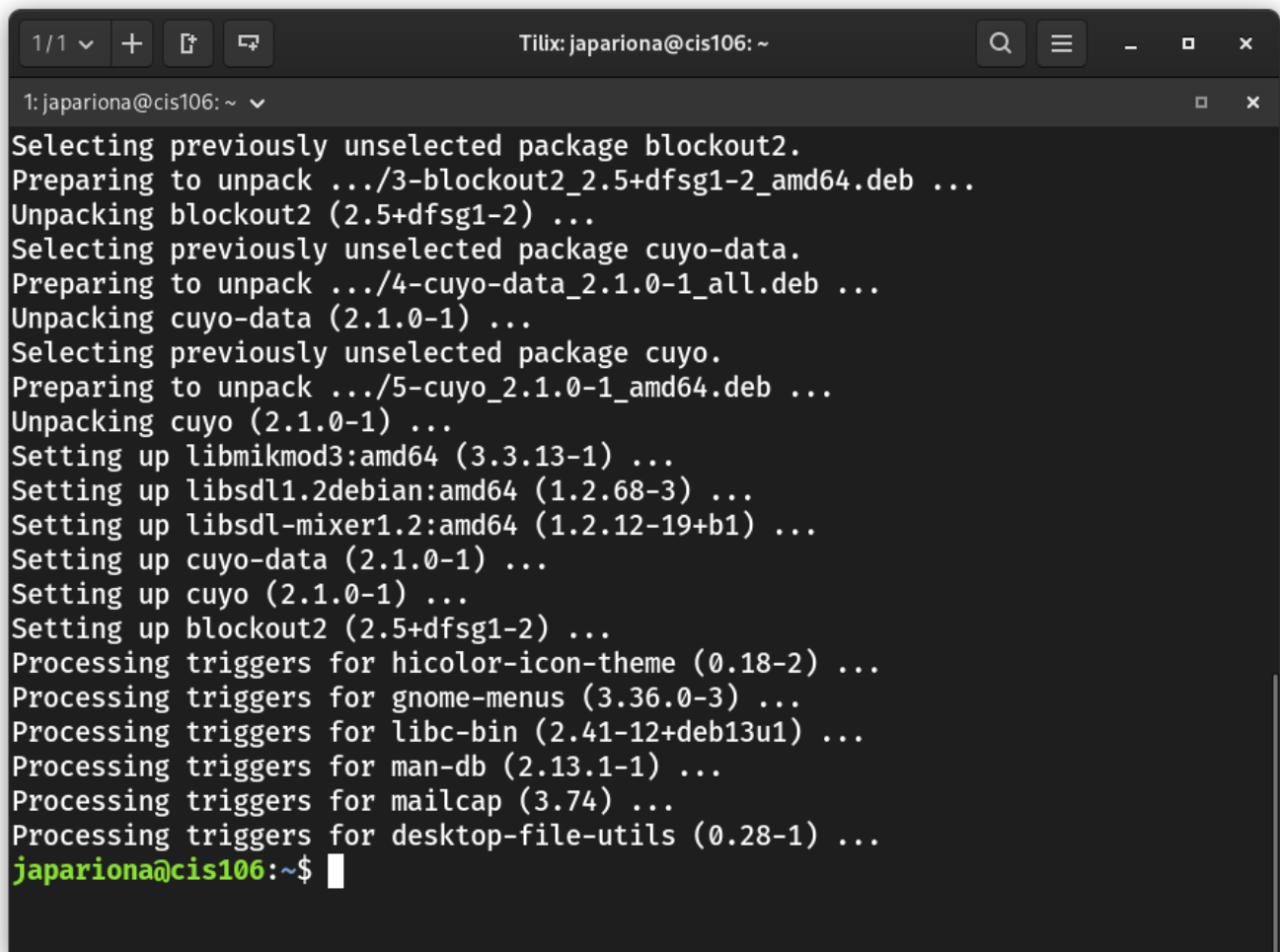
Lab 4 Submission

Question 1



The screenshot shows a terminal window with three panes. The left pane shows the installation progress for `ncurses`, `ttris`, and `tint`. The middle pane shows the installation progress for `blockattack`, `blockout2`, and `blocks-of-the-undead-data`. The right pane shows the installation progress for `games-tetris` and `vitetris`. The progress bars are at 100% for all packages.

```
1: japariona@cis106: ~  
[ncurses/stable 0.43.2-2 amd64  
ncurses Tetris clone with a bastard algorithm  
[ttris/stable 2.0.3-1 amd64  
very polished Tetris clone with CPU opponents  
[tint/stable 0.07 amd64  
Tetris clone for text based terminal  
-- MOST: *stdin* (1,1)100%  
2: japariona@cis106: ~  
[blockattack/stable 2.9.0-1+b1 amd64  
puzzle game inspired by Tetris  
[blockout2/stable 2.5+dfsg1-2 amd64  
Tetris like game (3D-tetris)  
[blocks-of-the-undead/stable 1.0-7 amd64  
Tetris Attack clone with spooky undertones  
[blocks-of-the-undead-data/stable 1.0-7 all  
-- MOST: *stdin* (1,1)0%  
3: japariona@cis106: ~  
[games-tetris/stable 5 all  
Debian's tetris-like games  
[vitetris/stable 0.59.1-3 amd64  
Virtual terminal *tris clone  
-- MOST: *stdin* (1,1)100%
```



The screenshot shows a terminal window with the output of the `dpkg` command. The output lists the packages to be installed and their dependencies, including `libmikmod3:amd64`, `libsdl1.2debian:amd64`, `libsdl-mixer1.2:amd64`, `cuyo-data`, `cuyo`, `blockout2`, `hicolor-icon-theme`, `gnome-menus`, `libc-bin`, `man-db`, `mailcap`, and `desktop-file-utils`.

```
1: japariona@cis106: ~  
Selecting previously unselected package blockout2.  
Preparing to unpack .../3-blockout2_2.5+dfsg1-2_amd64.deb ...  
Unpacking blockout2 (2.5+dfsg1-2) ...  
Selecting previously unselected package cuyo-data.  
Preparing to unpack .../4-cuyo-data_2.1.0-1_all.deb ...  
Unpacking cuyo-data (2.1.0-1) ...  
Selecting previously unselected package cuyo.  
Preparing to unpack .../5-cuyo_2.1.0-1_amd64.deb ...  
Unpacking cuyo (2.1.0-1) ...  
Setting up libmikmod3:amd64 (3.3.13-1) ...  
Setting up libsdl1.2debian:amd64 (1.2.68-3) ...  
Setting up libsdl-mixer1.2:amd64 (1.2.12-19+b1) ...  
Setting up cuyo-data (2.1.0-1) ...  
Setting up cuyo (2.1.0-1) ...  
Setting up blockout2 (2.5+dfsg1-2) ...  
Processing triggers for hicolor-icon-theme (0.18-2) ...  
Processing triggers for gnome-menus (3.36.0-3) ...  
Processing triggers for libc-bin (2.41-12+deb13u1) ...  
Processing triggers for man-db (2.13.1-1) ...  
Processing triggers for mailcap (3.74) ...  
Processing triggers for desktop-file-utils (0.28-1) ...  
japariona@cis106:~$
```

```
1/1 v + [T] [C] Tilix: japariona@cis106: ~ [Q] [≡] [-] [□] [X]
1: japariona@cis106: ~ [□] [X]
Removing cuyo (2.1.0-1) ...
Processing triggers for hicolor-icon-theme (0.18-2) ...
Processing triggers for gnome-menus (3.36.0-3) ...
Processing triggers for mailcap (3.74) ...
Processing triggers for desktop-file-utils (0.28-1) ...
japariona@cis106:~$ sudo apt purge cuyo
The following packages were automatically installed and are no longer required:
  cuyo-data libopusfile0 libsdl2-image-2.0-0 libsdl2-ttf-2.0-0
  libfmt10 libphysfs1 libsdl2-mixer-2.0-0 libxmp4
Use 'sudo apt autoremove' to remove them.

REMOVING:
  cuyo*

Summary:
  Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 4
  Space needed: 0 B / 36.4 GB available

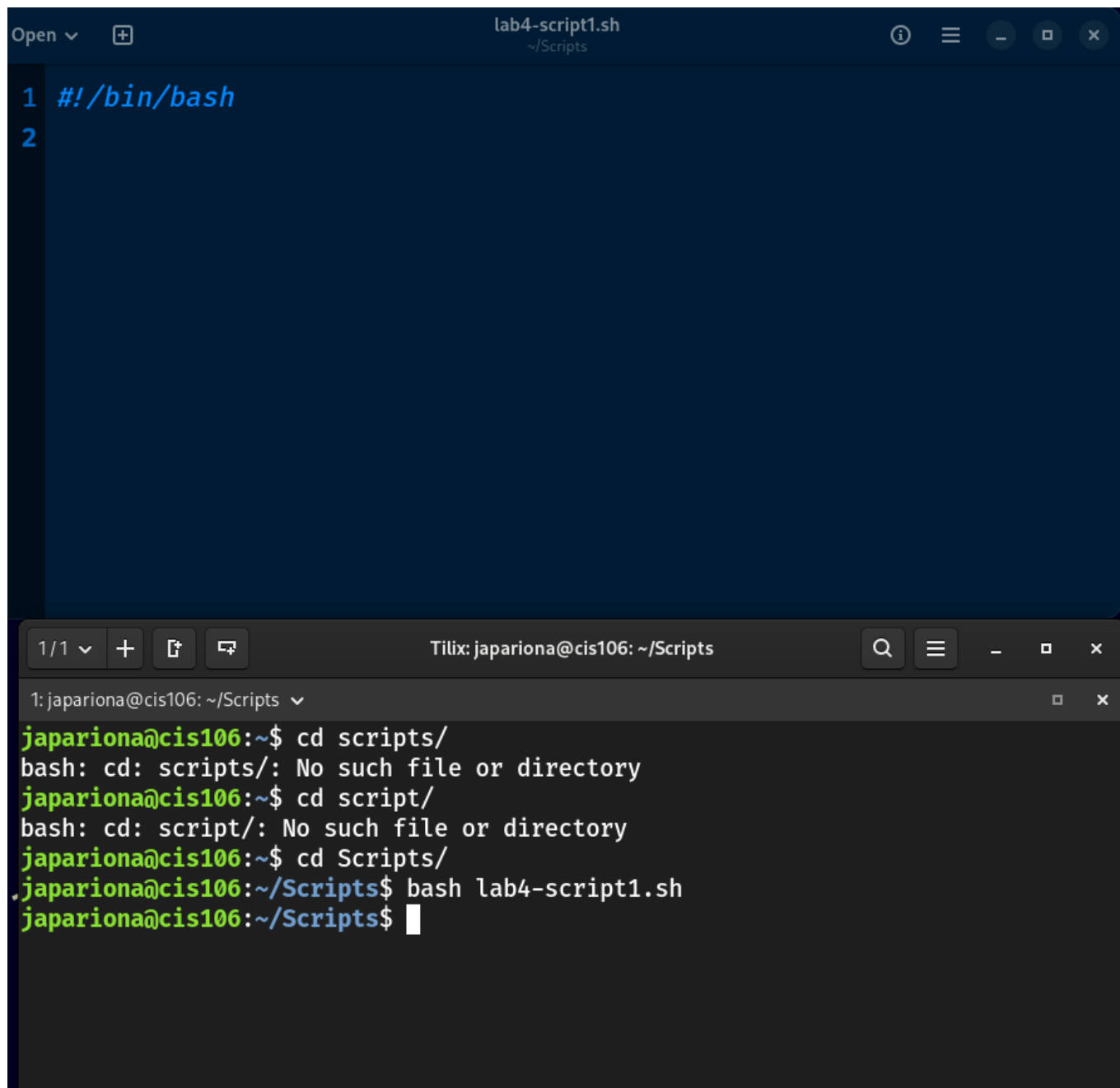
Continue? [Y/n] y
(Reading database ... 202215 files and directories currently installed.)
Purging configuration files for cuyo (2.1.0-1) ...
japariona@cis106:~$
```

```
Tilix: japariona@cis106: ~
1: japariona@cis106: ~
REMOVING:
  cuyo-data  libopusfile0  libsdl2-image-2.0-0  libsdl2-ttf-2.0-0
  libfmt10   libphysfs1    libsdl2-mixer-2.0-0  libxmp4

Summary:
  Upgrading: 0, Installing: 0, Removing: 8, Not Upgrading: 4
  Freed space: 7,495 kB

Continue? [Y/n] Y
(Reading database ... 202215 files and directories currently installed.)
Removing cuyo-data (2.1.0-1) ...
Removing libfmt10:amd64 (10.1.1+ds1-4) ...
Removing libsdl2-mixer-2.0-0:amd64 (2.8.1+dfsg-2) ...
Removing libopusfile0:amd64 (0.12-4+b3) ...
Removing libphysfs1:amd64 (3.0.2-6+b2) ...
Removing libsdl2-image-2.0-0:amd64 (2.8.8+dfsg-1) ...
Removing libsdl2-ttf-2.0-0:amd64 (2.24.0+dfsg-2) ...
Removing libxmp4:amd64 (4.6.2-2) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for libc-bin (2.41-12+deb13u1) ...
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 4
japariona@cis106:~$
```

Question 2

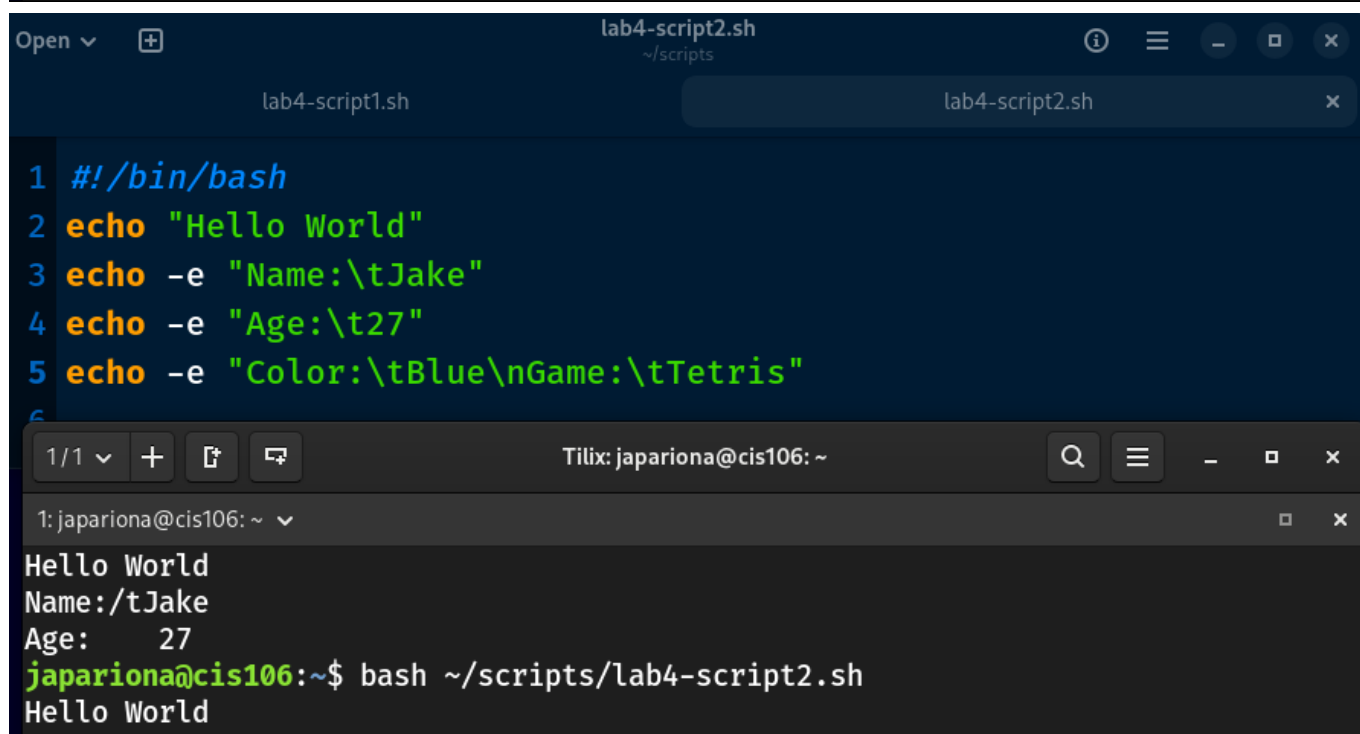


The screenshot shows a terminal window titled "lab4-script1.sh" with the path "~/Scripts". The script content is as follows:

```
1 #!/bin/bash
2
```

Below the script editor, a terminal window titled "Tilix: japariona@cis106: ~/Scripts" shows the following commands and output:

```
1: japariona@cis106: ~/Scripts
japariona@cis106:~$ cd scripts/
bash: cd: scripts/: No such file or directory
japariona@cis106:~$ cd script/
bash: cd: script/: No such file or directory
japariona@cis106:~$ cd Scripts/
japariona@cis106:~/Scripts$ bash lab4-script1.sh
japariona@cis106:~/Scripts$
```



The screenshot shows a terminal window titled "lab4-script2.sh" with the path "~/scripts". The script content is as follows:

```
1 #!/bin/bash
2 echo "Hello World"
3 echo -e "Name:\tJake"
4 echo -e "Age:\t27"
5 echo -e "Color:\tBlue\nGame:\tTetris"
6
```

Below the script editor, a terminal window titled "Tilix: japariona@cis106: ~" shows the following commands and output:

```
1: japariona@cis106: ~
Hello World
Name:/tJake
Age: 27
japariona@cis106:~$ bash ~/scripts/lab4-script2.sh
Hello World
```

```
Name: Jake
Age: 27
japariona@cis106:~$ bash ~/scripts/lab4-script2.sh
Hello World
Name: Jake
Age: 27
Color: Blue
Game:\Tetris
japariona@cis106:~$ bash ~/scripts/lab4-script2.sh
Hello World
Name: Jake
Age: 27
Color: Blue
Game: Tetris
japariona@cis106:~$
```

```
Open ▾ + lab4-script3.sh ~/scripts
1 #!/bin/bash
2 echo "Today's date is: "
3 date
4 echo -n "The current user is: "
5 whoami
6 echo "Memory Usage: "
7 free -h
```

Tilix: japariona@cis106: ~

1: japariona@cis106: ~ ▾

```
japariona@cis106:~$ bash ~/scripts/lab4-script3.sh
Today's date is:
Fri Mar 6 11:48:21 PM EST 2026
japariona@cis106:~$ bash ~/scripts/lab4-script3.sh
Today's date is:
Fri Mar 6 11:51:33 PM EST 2026
The current user is: japariona
japariona@cis106:~$ bash ~/scripts/lab4-script3.sh
Today's date is:
Fri Mar 6 11:54:05 PM EST 2026
The current user is: japariona
Memory Usage:

```

	total	used	free	shared	buff/cache	available
Mem:	3.8Gi	2.4Gi	493Mi	35Mi	1.2Gi	1.5Gi
Swap:	2.6Gi	0B	2.6Gi			

```
japariona@cis106:~$
```

Question 3

challenge_lab4.sh
~/cis106/labs/lab4

1 #!/bin/bash
2 echo "
3 figlet System
4 echo "DATE UTC: \$(date -u +"%m/%d/%y %H:%M:%S")"
5 echo "HOSTNAME: \$(hostname)"
6 echo "UPTIME: \$(uptime -p)"
7 echo "USER: \$(whoami)"
8 echo "
9 figlet MEMORY
10 free -h
11 echo "
12 figlet DISK
13 df -h /
14 echo "
15
16
17
18
19
20
21
22
23

Tilix: japariona@cis106: ~

1: japariona@cis106: ~

System
DATE UTC: 03/08/26 04:11:10
HOSTNAME: cis106
UPTIME: up 2 hours, 45 minutes
USER: japariona

MEMORY
Mem: total used free shared buff/cache available
Swap: 2.6Gi 0B 2.6Gi 36Mi 1.3Gi 1.4Gi

DISK
Filesystem Size Used Avail Use% Mounted on
/dev/sda1 47G 9.5G 35G 22% /

japariona@cis106:~\$

script